

UNIVERSITY OF WYOMING

MATH/COSC 4340: Numerical Methods for Differential Equations

Spring 2024

Tuesday/Thursday, 2:45 PM-4:00 PM, 105 Classroom Building

Instructor: Dane Taylor, dane.taylor@uwyo.edu

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Adjunct Professor, Department of Electrical Engineering and Computer Science

Contact information: Please direct all communication through WyoCourses.

Office hours: Wednesday 11:00 AM-12:00 and Wednesday 8:30 AM-9:30 AM in office EN 4087.

Course prerequisites: MATH 2310 and MATH 3340 (or instructor approval).

Course Description: Further develops the skills needed for computational problem solving and numerical analysis. Topics addressed include the following: one-step and linear multistep methods for solving initial value problems; truncation errors, stability analysis, and convergence of the numerical methods; finite difference approximation for elliptic equations and initial boundary value problems; numerical methods for sparse linear systems. Importantly, the course requires extensive coding using the Python programming language and Jupyter notebooks, which should be installed using Anaconda.

Student Learning Outcomes: Students will develop technical skills such as programming and mathematics. Students will learn how to approximately solve different classes of differential equations using a computer, while simultaneously accounting for the so-introduced approximation error. They should understand a variety of algorithmic approaches as well as the theoretical foundations supporting their derivations. Students should be able to communicate effectively in a technical and discipline specific context, both in writing and orally.

The student learning outcomes addressed in this course at a development or proficient level are: Apply computational thinking skills in a variety of contexts (problem solving; logic). Apply computing in discipline specific contexts. Build breadth and depth sequencing for more complex understanding. Demonstrate competency in coding to solve a problem. Differentiate Human/Computer element (consequences; modeling). Preparing for career/future education (verification).

Required texts, readings, and special tools or materials: No required text. A PDF of an optional textbook will be made available for download: "Numerical Analysis, 2nd edition", by Timothy Sauer.

General requirements and expectations for the course: It is expected that students will attend all classes and submit all assignments on time. Students should complete all assignments to stay current with course content. If students are going to be absent or need to submit assignments after due dates, they must communicate that with the instructor. Students should actively participate in class discussions and activities to gain the most out of the course. Students should demonstrate professionalism in their behavior, dress, and communication. Students should show respect for their peers, instructors, and the learning environment. Students should adhere to all academic integrity policies and procedures.

Grading Scale and Grading Policies: The grading scale and policies used in this course are:
A (90-100), B-(80-89), C (70-79), D (60-69).

Assignments and exams will be graded on a 100-point scale. Participation and attendance will also be considered when determining the final grade. All assignments must be completed and submitted on time to receive full credit. Late submissions will be subject to a penalty of up to 50% points deducted. Make-up exams will only be permitted in the case of an excused absence. Cheating and plagiarism will not be tolerated and will result in a zero for the assignment.

Required examinations, assignments, activities: Homework will be assigned weekly and will require answering both written and computational problems using the Python programming language. Assignments must be uploaded to WyoCourses. Grades will be weighted according to:

44%	Homework
22%	Exam 1
22%	Exam 2
6%	Quizzes
6%	Class participation (in class and online)

There will be two hand-written exams with the tentative dates:

Exam 1 on Thursday March 16 (week 7, before spring break)

Exam 2 on Thursday May 11 (finals week)

Important Dates:

Tue., Jan. 16 - Class start

Sat., Jan. 20 - Last day to add or change class enrollment

Th., Jan. 25 - Last day to drop class

March 11-15 - Spring break

Fri., Apr. 5 - Last day to withdraw

Th., May 2 - Last class

Required Participation Outside of Class Meetings: Students are encouraged to learn about and engage with computing resources around campus and the state including but not limited to [Advanced Research Computing Center \(ARCC\)](#); [Center for Blockchain and Digital Innovation](#); [Coe Student Innovation Center](#); [Data Science Center](#); [Digital Scholarship Center](#); [Innovation Wyrkshop](#); [NCAR Wyoming Supercomputing Center \(NWSC\)](#); [NSF EPSCOR](#); [Shell 3D Visualization Center](#); [Wyoming Geographic Information Science Center \(WYGISC\)](#); [Wyoming NIH INBRE](#); [Wyoming Water Resources Data System \(WRDS\)](#).

Attendance and Absence policies. Regular, full attendance is expected of all students. Students should arrive to class on time and be prepared for each class. Absences should be reported to the professor or instructor as soon as possible. Class participation is encouraged and is an important part of the learning process. Excessive absences may result in a lower grade or a failure of the course. This course follows the University policies on excused absences as outlined in [UW Regulation 2-108 \(Student Attendance Policy\)](#).

Classroom Behavior Policy: Always, treat your presence in the classroom and your enrollment in this course as you would a job. Act professionally, arrive on time, pay attention, complete your work in a timely and professional manner, and treat all deadlines seriously. You will be respectful

towards you classmates and instructor. Spirited debate and disagreement are to be expected in any classroom and all views will be heard fully, but always we will behave civilly and with respect towards one another. Personal attacks, offensive language, name-calling, and dismissive gestures are not warranted in a learning atmosphere. As the instructor, I have the right to dismiss you from the classroom, study sessions, electronic forums, and other areas where disruptive behavior occurs.

Electronic devices such as mobile phones and laptops should be used as tools for learning. These devices are allowed for research, note-taking, and other academic purposes; those who require laptops for accommodation of disabilities, work with Disability Support Services and me to accommodate your needs. No unauthorized video or audio recording during class is allowed to protect the privacy of your fellow students; for those who require recording for accommodation of disabilities, work with Disability Support Services and me to accommodate your needs.

Classroom Statement on Diversity: The University of Wyoming values an educational environment that is diverse, equitable, and inclusive. The diversity that students and faculty bring to class, including age, country of origin, culture, disability, economic class, ethnicity, gender identity, immigration status, linguistic, political affiliation, race, religion, sexual orientation, veteran status, worldview, and other social and cultural diversity is valued, respected, and considered a resource for learning.

Disability Support: The University of Wyoming is committed to providing equitable access to learning opportunities for all students. If you have a disability, including but not limited to physical, learning, sensory or psychological disabilities, and would like to request accommodations in this course due to your disability, please register with and provide documentation of your disability as soon as possible to Disability Support Services (DSS), Room 128 Knight Hall. You may also contact DSS at (307) 766-3073 or udss@uwyo.edu. It is in the student's best interest to request accommodations within the first week of classes, understanding that accommodations are not retroactive. Visit the DSS website for more information at: www.uwyo.edu/udss

Academic Dishonesty Policies: Academic dishonesty will not be tolerated in this class. Cases of academic dishonesty will be treated in accordance with UW Regulation 2-114. The penalties for academic dishonesty can include, at my discretion, an "F" on an exam, an "F" on the class component exercise, and/or an "F" in the entire course. Academic dishonesty means anything that represents someone else's ideas as your own without attribution. It is intellectual theft – stealing - and includes (but is not limited to) unapproved assistance on examinations, plagiarism (use of any amount of another person's writings, blog posts, publications, and other materials without attributing that material to that person with citations), or fabrication of referenced information. Facilitation of another person's academic dishonesty is also considered academic dishonesty and will be treated identically.

Duty to Report: UW faculty are committed to supporting students and upholding the University's non-discrimination policy. Under Title IX, discrimination based upon sex and gender is prohibited. If you experience an incident of sex- or gender-based discrimination, we encourage you to report it. While you may talk to a faculty member, understand that as a "Responsible Employee" of the University, the faculty member MUST report information you share about the incident to the university's Title IX Coordinator (you may choose whether you or anyone involved is identified by name). If you would like to speak with someone who may be able to afford you privacy or

confidentiality, there are people who can meet with you. Faculty can help direct you or you may find info about UW policy and resources at <http://www.uwyo.edu/reportit>

You do not have to go through the experience alone. Assistance and resources are available, and you are not required to make a formal complaint or participate in an investigation to access them.

Substantive changes to syllabus: Note: This syllabus is a guide. Circumstances may alter the reading and/or test schedules. You are required to check WyoCourses at least once a week.

Schedule:

Week 1-4 - ODEs: Initial Value Problems

- Introduction and Truncated Taylor Expansions
- Forward Euler, Trapezoidal and Taylor Methods
- Systems of ODEs and Runge-Kutta Methods
- Backward Euler, Implicit Methods and Stiff Equations
- Multistep Methods and their Stability
- Absolute Stability and A Stability

Week 5-8 - ODEs: Boundary Value Problems

- Shooting Methods
- Finite Difference Methods
- Collocation and Finite Element Methods

Week 9-11 – Numerical Linear Algebra

- Dense Linear Solvers
- Computational Complexity
- Iterative Methods
- Sparse and Low-Rank Methods
- Spectral Solutions to ODEs
- Numerical Methods for Spectral Decompositions

Week 12-15 – Finite Difference Methods for PDEs

- Parabolic Equations and the Heat Equation
- Hyperbolic Equations and the Wave Equation
- Elliptic Equations
- Nonlinear PDEs

Campus Resources:

DISABILITY SUPPORT SERVICES: udss@uwyo.edu, 766-3073, 128 Knight Hall, www.uwyo.edu/udss

COUNSELING CENTER: uccstaff@uwyo.edu, 766-2187, 766-8989 (After hours), 341 Knight Hall, www.uwyo.edu/ucc

ACADEMIC AFFAIRS: 766-4286, 312 Old Main, www.uwyo.edu/acadaffairs

DEAN OF STUDENTS OFFICE: dos@uwyo.edu, 766-3296, 128 Knight Hall, www.uwyo.edu/dos

UW POLICE DEPARTMENT: uwupd@uwyo.edu, 766-5179, 1426 E Flint St, www.uwyo.edu/uwpd

STUDENT CODE OF CONDUCT WEBSITE: www.uwyo.edu/dos/conduct